

Complex Numbers and Trigonometry — Revision

Summary

Algebra, geometry, identities, and the unifying power of $e^{i\theta}$

1 Complex numbers — the basics

A complex number $z = x + iy$ with $x, y \in \mathbb{R}$ and $i^2 = -1$. Identify $\mathbb{C}$ with the plane $\mathbb{R}^2$ via $z \leftrightarrow (x, y)$.

1.1 Three forms of z

- **Cartesian:** $z = x + iy$. Real part $\operatorname{Re}(z) = x$, imaginary part $\operatorname{Im}(z) = y$.
- **Polar:** $z = r(\cos \theta + i \sin \theta)$ where $r = |z| = \sqrt{x^2 + y^2}$, $\theta = \arg z = \arctan(y/x)$ (with quadrant care).
- **Exponential (Euler):** $z = re^{i\theta}$, using $e^{i\theta} = \cos \theta + i \sin \theta$.

Conjugate. $\bar{z} = x - iy = re^{-i\theta}$. Properties: $z + \bar{z} = 2\operatorname{Re}(z)$, $z - \bar{z} = 2i\operatorname{Im}(z)$, $z\bar{z} = |z|^2$. Conjugation distributes: $\overline{z_1 \pm z_2} = \bar{z}_1 \pm \bar{z}_2$, $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$, $\overline{z_1/z_2} = \bar{z}_1/\bar{z}_2$.

Modulus and argument under operations.

- Multiplication: $|z_1 z_2| = |z_1||z_2|$, $\arg(z_1 z_2) = \arg z_1 + \arg z_2 \pmod{2\pi}$.
- Division: $|z_1/z_2| = |z_1|/|z_2|$, $\arg(z_1/z_2) = \arg z_1 - \arg z_2$.
- Addition: only $|z_1 + z_2| \leq |z_1| + |z_2|$ (triangle inequality), no clean argument formula.

1.2 Operations as geometry

- **Addition** $z + w$: vector sum (parallelogram rule).
- **Multiplication by $w = \rho e^{i\phi}$** : scale by ρ and rotate by ϕ anticlockwise. In particular, multiplication by i is a 90° anticlockwise rotation.
- **Conjugation**: reflection across the real axis.
- **Inversion** $z \mapsto 1/z$: $1/z = \bar{z}/|z|^2$, so it's reflect-and-scale-by- $1/|z|^2$.

1.3 Useful identities

- **Parallelogram identity:** $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$.
- **Triangle inequality:** $||z_1| - |z_2|| \leq |z_1 \pm z_2| \leq |z_1| + |z_2|$.
- **Reverse triangle:** $|z - w| \geq ||z| - |w||$.
- **Lagrange identity:** $|z_1|^2 |z_2|^2 - |\operatorname{Re}(z_1 \bar{z}_2)|^2 = |\operatorname{Im}(z_1 \bar{z}_2)|^2$.
- $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + 2\operatorname{Re}(z_1 \bar{z}_2)$ (the “cosine rule”).

2 De Moivre and roots of unity

Theorem (De Moivre). $(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$ for all $n \in \mathbb{Z}$.

In Euler form this is just $(e^{i\theta})^n = e^{in\theta}$, a single line.

2.1 n th roots of a complex number

The n th roots of $w = \rho e^{i\phi}$ are

$$z_k = \rho^{1/n} e^{i(\phi + 2\pi k)/n}, \quad k = 0, 1, \dots, n-1.$$

They sit on a circle of radius $\rho^{1/n}$ at angles equally spaced by $2\pi/n$, forming a regular n -gon.

2.2 n th roots of unity

Setting $w = 1$ gives the n th roots of unity:

$$\omega_k = e^{2\pi i k/n}, \quad k = 0, 1, \dots, n-1.$$

Write $\omega = e^{2\pi i/n}$ for the *primitive* n th root; then $\omega_k = \omega^k$.

Key identities.

- **Sum:** $1 + \omega + \omega^2 + \dots + \omega^{n-1} = 0$ for $n \geq 2$. (Geometric series; or: roots of $z^n - 1 = (z-1)(z^{n-1} + \dots + 1)$.)
- **Product:** $\prod_{k=0}^{n-1} \omega_k = (-1)^{n-1}$ (constant term of $z^n - 1$).
- **Polynomial factorisation:** $z^n - 1 = \prod_{k=0}^{n-1} (z - \omega^k)$.
- **Cyclotomic factorisation:** $z^n - 1 = \prod_{d|n} \Phi_d(z)$, where Φ_d is the d th cyclotomic polynomial. Most-used: $\Phi_1 = z - 1$, $\Phi_2 = z + 1$, $\Phi_3 = z^2 + z + 1$, $\Phi_4 = z^2 + 1$, $\Phi_p = z^{p-1} + \dots + 1$ for prime p .

Filtering by congruence (root-of-unity filter). For polynomial $P(z) = \sum a_k z^k$:

$$\sum_{k \equiv r \pmod{n}} a_k = \frac{1}{n} \sum_{j=0}^{n-1} \omega^{-rj} P(\omega^j).$$

Standard application: extract the sum of every third binomial coefficient $\binom{n}{0} + \binom{n}{3} + \binom{n}{6} + \dots$ from $(1+z)^n$ at the cube roots of unity.

Sums of cosines/sines of equally spaced angles. Take real/imaginary parts of $\sum \omega^k = 0$:

$$\sum_{k=0}^{n-1} \cos\left(\frac{2\pi k}{n}\right) = 0, \quad \sum_{k=0}^{n-1} \sin\left(\frac{2\pi k}{n}\right) = 0 \quad (n \geq 2).$$

3 Geometry in the complex plane

Three operations cover most plane-geometry problems:

Operation	Complex form
Translate by w	$z \mapsto z + w$
Rotate by θ about origin (anti-clockwise)	$z \mapsto e^{i\theta} z$
Rotate by θ about z_0 (anticlockwise)	$z \mapsto z_0 + e^{i\theta}(z - z_0)$
Scale by r about origin	$z \mapsto rz$
Reflect across real axis	$z \mapsto \bar{z}$
Reflect across line through origin at angle α	$z \mapsto e^{2i\alpha} \bar{z}$
Spiral similarity (rotate θ anti-clockwise, scale r)	$z \mapsto re^{i\theta} z$

Convention: positive θ rotates anticlockwise (the mathematical convention, matching $e^{i\theta}$ tracing the unit circle anticlockwise as θ increases). For a clockwise rotation by θ , use $e^{-i\theta}$ instead.

3.1 Standard criteria

Collinearity. Three points z_1, z_2, z_3 are collinear iff $\frac{z_3 - z_1}{z_2 - z_1} \in \mathbb{R}$, i.e. the ratio is its own conjugate.

Concyclicity (or collinearity). Four points z_1, z_2, z_3, z_4 lie on a common circle (or line) iff the cross-ratio

$$(z_1, z_2; z_3, z_4) = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)} \in \mathbb{R}.$$

Equilateral triangle. Three points z_1, z_2, z_3 form an equilateral triangle iff

$$z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1,$$

or equivalently $z_1 + \omega z_2 + \omega^2 z_3 = 0$ for one of the cube roots of unity ω .

Right angle at z_2 . The angle $\angle z_1 z_2 z_3 = 90^\circ$ iff $\frac{z_3 - z_2}{z_1 - z_2}$ is purely imaginary.

Centroid, circumcentre. Centroid of triangle: $(z_1 + z_2 + z_3)/3$. Circumcentre on the unit circle (when $|z_i| = 1$): 0. For three points on the unit circle, $z_1 + z_2 + z_3$ is the orthocentre.

4 Trigonometric identities via $e^{i\theta}$

The Euler formula $e^{i\theta} = \cos \theta + i \sin \theta$ converts trigonometric manipulations into algebra of exponentials. This is the unifying theme of this whole topic.

4.1 Basic identities, derived in one line

- $\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$
- **Sum-of-angles:** $e^{i(\alpha+\beta)} = e^{i\alpha} e^{i\beta}$. Expand to get $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$ and $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$.

- **Double angle:** $\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$, $\sin 2\theta = 2 \sin \theta \cos \theta$.
- **Multiple angle:** $e^{in\theta} = (\cos \theta + i \sin \theta)^n$. Expanding by the binomial theorem and matching real/imaginary parts gives $\cos(n\theta)$ and $\sin(n\theta)$ as polynomials in $\cos \theta, \sin \theta$.

Standard multiple-angle formulas.

- $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$
- $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$
- $\cos 4\theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1$

4.2 Sum-to-product and product-to-sum

Multiply $e^{i\alpha} \cdot e^{i\beta} = e^{i(\alpha+\beta)}$ or add $e^{i\alpha} + e^{i\beta}$ and factor $e^{i(\alpha+\beta)/2}$:

- $\sin \alpha + \sin \beta = 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$
- $\sin \alpha - \sin \beta = 2 \cos \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$
- $\cos \alpha + \cos \beta = 2 \cos \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$
- $\cos \alpha - \cos \beta = -2 \sin \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$
- $2 \sin \alpha \cos \beta = \sin(\alpha + \beta) + \sin(\alpha - \beta)$
- $2 \cos \alpha \cos \beta = \cos(\alpha + \beta) + \cos(\alpha - \beta)$
- $2 \sin \alpha \sin \beta = \cos(\alpha - \beta) - \cos(\alpha + \beta)$

4.3 Chebyshev polynomials — $\cos(n\theta)$ as a polynomial in $\cos \theta$

There exists a unique polynomial T_n such that $\cos(n\theta) = T_n(\cos \theta)$. These are the *Chebyshev polynomials of the first kind*:

$$T_0 = 1, \quad T_1 = x, \quad T_2 = 2x^2 - 1, \quad T_3 = 4x^3 - 3x, \quad T_4 = 8x^4 - 8x^2 + 1, \dots$$

Recurrence: $T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$.

Why useful. Anything of the form “ $\cos \theta$ satisfies a polynomial equation” becomes accessible. Roots of $T_n(x) = 0$ are $\cos\left(\frac{(2k-1)\pi}{2n}\right)$ for $k = 1, \dots, n$. Roots of $T_n(x) = 1$ include $\cos(2\pi k/n)$.

Sister polynomial. U_{n-1} such that $\sin(n\theta) = \sin \theta \cdot U_{n-1}(\cos \theta)$ (Chebyshev of the second kind).

5 Sum identities — the geometric-series workhorse

When you see $\sum \cos(k\theta)$ or $\sum \sin(k\theta)$, write it as Re/Im of a geometric series:

$$\sum_{k=0}^{n-1} e^{ik\theta} = \frac{e^{in\theta} - 1}{e^{i\theta} - 1} = \frac{\sin(n\theta/2)}{\sin(\theta/2)} \cdot e^{i(n-1)\theta/2}.$$

Standard consequences.

- $\sum_{k=0}^{n-1} \cos(k\theta) = \frac{\sin(n\theta/2)}{\sin(\theta/2)} \cos\left(\frac{(n-1)\theta}{2}\right).$
- $\sum_{k=0}^{n-1} \sin(k\theta) = \frac{\sin(n\theta/2)}{\sin(\theta/2)} \sin\left(\frac{(n-1)\theta}{2}\right).$
- **Lagrange’s trigonometric identity:** $1 + \sum_{k=1}^n \cos(k\theta) = \frac{1}{2} + \frac{\sin((2n+1)\theta/2)}{2 \sin(\theta/2)}.$

Roots-of-unity sums of $\cos$ and $\sin$. Setting $\theta = 2\pi/n$ in the geometric series:

$$\sum_{k=0}^{n-1} \cos\left(\frac{2\pi k}{n}\right) = 0, \quad \sum_{k=0}^{n-1} \sin\left(\frac{2\pi k}{n}\right) = 0 \quad (n \geq 2).$$

Removing the $k = 0$ term: $\sum_{k=1}^{n-1} \cos(2\pi k/n) = -1$.

Lacunary sums (every m th term). Use root-of-unity filter. For example, sum of $\cos(k\theta)$ over odd k equals half of the difference between $\sum_k \cos(k\theta)$ and $\sum_k (-1)^k \cos(k\theta) = \sum_k \cos(k(\theta + \pi))$.

6 Classical algebraic-trigonometric identities

6.1 Special angle values

$$\cos(\pi/5) = \frac{1 + \sqrt{5}}{4}$$

$$\cos(\pi/3) = 1/2$$

$$\cos(\pi/6) = \sqrt{3}/2$$

$$\sin(\pi/10) = \frac{\sqrt{5} - 1}{4}$$

$$\cos(2\pi/5) = \frac{\sqrt{5} - 1}{4}$$

$$\cos(\pi/4) = \sqrt{2}/2$$

$$\cos(\pi/12) = \frac{\sqrt{6} + \sqrt{2}}{4}$$

$$\sin(3\pi/10) = \frac{\sqrt{5} + 1}{4}$$

The values at $\pi/5$ come from $\cos(2\pi/5)$ being a root of $4x^2 + 2x - 1 = 0$ (from the cyclotomic polynomial $\Phi_5(z)$ when $z = e^{2\pi i/5}$ is reduced to a quadratic in $z + z^{-1} = 2\cos(2\pi/5)$).

6.2 Product identities

Product of sines at equally spaced angles.

$$\prod_{k=1}^{n-1} \sin\left(\frac{k\pi}{n}\right) = \frac{n}{2^{n-1}}.$$

Examples.

- $n = 3$: $\sin(\pi/3)\sin(2\pi/3) = (\sqrt{3}/2)^2 = 3/4 = 3/2^2$. ✓
- $n = 5$: $\prod_{k=1}^4 \sin(k\pi/5) = 5/16$.
- $n = 7$: $\prod_{k=1}^6 \sin(k\pi/7) = 7/64$.

Cosine analogue.

$$\prod_{k=1}^{n-1} 2\sin\left(\frac{k\pi}{n}\right) = n.$$

This follows from factoring $z^n - 1 = \prod (z - e^{2\pi i k/n})$ at $z = 1$ and taking moduli.

6.3 Cyclotomic-polynomial-style identities

- $z^n - 1 = (z - 1)(z^{n-1} + z^{n-2} + \dots + 1)$.
- $z^n + 1 = \prod_{k=0}^{n-1} (z - e^{i(2k+1)\pi/n})$ (the $2n$ th roots of unity that aren't n th roots).
- $z^4 + z^2 + 1 = (z^2 + z + 1)(z^2 - z + 1)$.
- $z^5 - 1 = (z - 1)(z^2 - 2\cos(2\pi/5)z + 1)(z^2 - 2\cos(4\pi/5)z + 1)$.

6.4 Algebraicity of $\cos(2\pi/n)$

For every n , $\cos(2\pi/n)$ is an algebraic number satisfying a polynomial of degree $\varphi(n)/2$ (where φ is Euler's totient), obtained from the minimal polynomial of $e^{2\pi i/n} + e^{-2\pi i/n}$. Constructive case: $\cos(2\pi/n)$ is constructible by ruler and compass iff n is a product of a power of 2 and distinct Fermat primes (Gauss–Wantzel).

7 Triangle, polygon, and locus problems

7.1 Triangle identities (in $\mathbb{R}$)

For any triangle with angles $A + B + C = \pi$:

- $\cos^2 A + \cos^2 B + \cos^2 C + 2\cos A \cos B \cos C = 1$.
- $\sin 2A + \sin 2B + \sin 2C = 4\sin A \sin B \sin C$.
- $\cos 2A + \cos 2B + \cos 2C = -1 - 4\cos A \cos B \cos C$.
- $\tan A + \tan B + \tan C = \tan A \tan B \tan C$ (since $A + B + C = \pi$ forces $\tan(A + B) = -\tan C$).

7.2 Inequalities for triangle angles

- $\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}$ with equality iff equilateral.
- $\cos A + \cos B + \cos C \leq 3/2$ (equality at equilateral).
- $\sin A \sin B \sin C \leq 3\sqrt{3}/8$ (equality at equilateral).

All these come from concavity (or convexity) of $\sin, \cos$ on $(0, \pi)$ via Jensen.

7.3 Locus problems

The unit circle as a workspace. If the problem says “points on the unit circle,” use $z\bar{z} = 1$, so $\bar{z} = 1/z$. Many identities collapse beautifully.

Standard loci.

- $|z - z_0| = r$: circle of radius r centred at z_0 .
- $|z - z_1| = |z - z_2|$: perpendicular bisector of segment from z_1 to z_2 .
- $|z - z_1|/|z - z_2| = k$ (constant $\neq 1$): *Apollonius circle*.
- $\arg\left(\frac{z - z_1}{z - z_2}\right) = \alpha$ (constant): an arc of a circle through z_1 and z_2 .
- $\operatorname{Re}\left(\frac{z - z_1}{z - z_2}\right) = 0$: the circle with diameter $z_1 z_2$ (right angle subtended).

8 Quick-reference cheat sheet

Pattern in the problem	Tool / trick to try first
Multiply complex numbers	Convert to polar; multiply moduli, add arguments
$z^n = w$	n th roots formula: $ w ^{1/n} e^{i(\arg w + 2\pi k)/n}$
Sum every m th coefficient of $(1 + z)^n$	Root-of-unity filter at m th roots of unity
Show z_1, z_2, z_3 collinear	$(z_3 - z_1)/(z_2 - z_1) \in \mathbb{R}$
Show four points concyclic	Cross-ratio $\in \mathbb{R}$
Equilateral triangle	$z_1 + \omega z_2 + \omega^2 z_3 = 0$
Right angle at z_2	$(z_3 - z_2)/(z_1 - z_2)$ purely imaginary
Rotation by θ about z_0 (anticlockwise)	$z \mapsto z_0 + e^{i\theta}(z - z_0)$
Reflection across line through 0 at angle α	$z \mapsto e^{2i\alpha} \bar{z}$
$\sum \cos(k\theta)$ or $\sum \sin(k\theta)$	Re/Im of geometric series in $e^{i\theta}$
Multiple angle $\cos(n\theta), \sin(n\theta)$	Expand $(\cos \theta + i \sin \theta)^n$ via binomial
$\cos(n\theta)$ as polynomial in $\cos \theta$	Chebyshev polynomial T_n
Product of $\sin(k\pi/n)$	$n/2^{n-1}$
Sum of $\cos(2\pi k/n)$, $k = 0, \dots, n-1$	Equals 0 for $n \geq 2$
$\sin \alpha \pm \sin \beta$, $\cos \alpha \pm \cos \beta$	Sum-to-product formulas
$\sin \alpha \sin \beta$, $\cos \alpha \cos \beta$, $\sin \alpha \cos \beta$	Product-to-sum formulas
$\cos(2\pi/5)$, $\cos(\pi/5)$, etc.	$(\sqrt{5} \pm 1)/4$, from Φ_5
$ z_1 + z_2 ^2$ from $ z_1 , z_2 $	Cosine rule: $ z_1 ^2 + z_2 ^2 + 2\operatorname{Re}(z_1 \bar{z}_2)$
$ z - z_1 / z - z_2 $ constant locus	Apollonius circle
$\arg((z - z_1)/(z - z_2))$ constant locus	Arc through z_1, z_2
Triangle identity $A + B + C = \pi$	$\tan A + \tan B + \tan C = \tan A \tan B \tan C$
Show $\cos \theta$ algebraic	Use Chebyshev polynomial for the right n

Three universal moves: (1) *Convert trig to exponential*. Whenever a problem mixes sines, cosines, sums, and products in awkward ways, replace by $e^{i\theta}$ and let algebra do the work. (2) *Recognise the geometric series*. Sums of $\cos(k\theta)$ or $\sin(k\theta)$ become $\sum e^{ik\theta}$, immediately summable. (3) *Use roots of unity for filtering*. Whenever you need to extract every m th term from a polynomial, reach for $\omega = e^{2\pi i/m}$ and the filter formula.